Focusing Influence Mechanism for Multi-Agent Reinforcement Learning

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Abstract

Cooperative multi-agent reinforcement learning (MARL) under sparse rewards remains fundamentally challenging because agents often fail to concentrate their influence, leading to insufficiently coordinated exploration. To address this, we propose the Focusing Influence Mechanism (FIM), a framework that encourages agents to focus their influence on under-explored parts of the state space through an entropy-based criterion, while leveraging eligibility traces to enable multiple agents to consistently align and sustain their influence on the same parts of the state space when beneficial, thereby promoting coordinated and persistent joint behavior. By emphasizing under-explored regions of the state space, FIM facilitates more efficient and structured exploration even under extremely sparse rewards. Across diverse MARL benchmarks, FIM consistently improves cooperative performance over strong baselines. Our implementation is available at https://unistmlic.github.io/FIM/.

1 Introduction

Cooperative multi-agent reinforcement learning (MARL) is a powerful framework for sequential decision-making problems involving multiple agents, with applications in autonomous driving [1], multi-robot coordination [2], and real-time strategy games [3]. These environments involve partial observability, making decentralized partially observable Markov decision processes (Dec-POMDPs) [4] a natural modeling choice. To address the challenges arising from limited observability, the centralized training with decentralized execution (CTDE) [5, 6] paradigm has been widely adopted. In CTDE, policies are trained using access to the global state and all agents' observations, but are executed using only local observations. Prominent CTDE methods such as VDN [7], QMIX [8], and QPLEX [9] leverage value decomposition to promote coordinated policy learning.

Despite the success of CTDE, many CTDE-based MARL methods struggle in sparse-reward settings where effective exploration is essential [10–12]. Several approaches have been proposed to address this challenge, including maximizing mutual influence between agents [11], prioritizing under-visited but important states [13], diversifying trajectories [14], and leveraging communication [15, 16]. While promising, we observe that these methods can still fail in challenging sparse-reward environments, particularly under extremely sparse rewards, because they do not explicitly guide how agents should coordinate and allocate their influence to escape local optima. Liu et al. [17] improve exploration efficiency by concentrating an agent's influence on an external state; however, selecting external-state dimensions relies on human knowledge and prior task information, which can be difficult to scale as the dimensionality of the external state increases or when such information is unavailable.

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To address these issues, we identify two factors that make sparse-reward MARL difficult. First, even when agents can influence the environment, their influence tends to remain scattered across multiple targets without converging on shared targets, leading to diluted and ineffective exploration. Second, agents lack a principled criterion for prioritizing parts of the state space that are under-explored relative to the current behavior policy. Motivated by these observations, we propose the **Focusing Influence Mechanism (FIM)**, a MARL framework that addresses both challenges through two complementary components: (i) an *agent focusing influence (AFI)* module that concentrates multiple agents' influence on shared targets by accumulating collective influence via eligibility traces and amplifying intrinsic rewards for components that receive consistent joint attention, thereby enabling persistent coordinated behavior, and (ii) a *state focusing influence (SFI)* module that uses an entropy-based criterion to prioritize under-explored parts of the state space, guiding AFI toward more meaningful targets. As a result, FIM promotes more efficient and structured exploration in challenging sparse-reward environments, and extensive experiments across diverse MARL benchmarks demonstrate improved performance over existing methods.

2 Related Works

Intrinsic Motivation for MARL. Intrinsic motivation is widely used to promote exploration in sparse-reward environments. Curiosity-driven objectives seek novel states [18, 13, 19–23], while trajectory diversity methods expand state-space coverage [24–27]. Committed exploration is induced via a shared latent variable [28, 29], and spatial formations reduce redundant exploration [30]. Subgoal-based methods decompose tasks into smaller objectives [31–34]. Exploration can also be focused in low-dimensional subspaces [12, 35, 36], and expectation alignment enables agents to adapt to anticipated peer behaviors [37].

Counterfactual Reasoning Based Credit Assignment. Counterfactual reasoning enables credit assignment by measuring each agent's contribution to the team reward. COMA estimates individual action advantages using counterfactual baselines [38–41], while predictive counterfactual models support value decomposition [42, 43]. Shapley value–based methods assign local credit by marginalizing individual contributions to the global reward [44–46]. In offline settings, counterfactual conservatism [47] and sample averaging [48] improve stability. Counterfactual reasoning also aids in identifying important agents [49] and salient states [50].

Influence-Driven Coordination. Influence-based methods aim to promote coordination by inducing causally significant changes. Social influence quantifies how an agent's actions affect the behaviors of its teammates [10, 51–54] and guides communication decisions [55]. Opponent modeling enables agents to influence policy updates of others [56–59]. Influence-aware exploration affects future dynamics [11, 60] or induces novel observations [61]. Influence has been extended to incentivize beneficial behaviors in others [62], discourage undesirable actions [63], or shape the returns of other agents [64], as well as to affect external states [17] or latent representations of the environment [65]. LAIES [17] measures the deviation of the realized transition from its counterfactual expectation, whereas FIM measures the influence on the magnitude of state change, directly capturing whether agents can enlarge the change of a target dimension. Our work builds on this line by aligning and concentrating influence across agents on key parts of the state space, enabling more persistent and structured coordination.

3 Preliminary

Decentralized POMDP and CTDE Setup. In MARL, the environment is typically modeled as a Dec-POMDP [4], defined by the tuple $\langle \mathcal{N}, S, A, P, R, O, \mathcal{O}, \gamma \rangle$, where $\mathcal{N}$ is a set of n agents, S is the global state space, $A = A^0 \times \cdots \times A^{n-1}$ is the joint action space, and γ is the discount factor. At each timestep t, each agent $i \in \mathcal{N}$ receives a local observation $o_t^i = \mathcal{O}(s_t, i)$ and chooses an action a_t^i from its policy π^i, based on its trajectory $\tau_t^i = (o_0^i, a_0^i, \ldots, o_t^i)$. The state s_t is defined as a D-dimensional vector, i.e., $s_t = (s_t^0, \cdots, s_t^{D-1})$, and for a given $(s_t, \mathbf{a}_t)$ pair, the environment transitions to $s_{t+1} \sim P(\cdot \mid s_t, \mathbf{a}_t)$ and returns a shared reward $r_t = R(s_t, \mathbf{a}_t)$. The goal is to learn a joint policy $\boldsymbol{\pi} = \prod_{i=1}^{n} \pi^i$ that maximizes the expected return $\sum_{t=0}^{\infty} \gamma^t r_t$. In this paper, we adopt the CTDE paradigm [8], where agents are trained using the global state to optimize a total value function Q^{tot}, while each agent executes actions based solely on local observations during deployment.

Credit Assignment via Counterfactual Reasoning. In the CTDE paradigm, credit assignment mechanisms [8, 47, 17] estimate each agent's contribution to team performance, supporting not only the optimization of a global value function but also promoting effective exploration [14], information sharing [30], and communication [66]. A widely adopted technique is counterfactual reasoning [38, 47, 17], which quantifies causal influence by comparing the actual outcome to a counterfactual one where only an individual agent's action is replaced. COMA [38], for example, defines credit for agent i as:

$$\text{credit}_t^i = f(s_t, \boldsymbol{\tau}_t, \mathbf{a}_t) - \mathbb{E}_{a_t^i \sim \pi^i(\cdot|o_t^i)} \left[f(s_t, \boldsymbol{\tau}_t, a_t^i, \mathbf{a}_t^{-i}) \right], \tag{1}$$

where $f = Q^{tot}$. This formulation has been leveraged not only for advantage estimation but also for shaping exploration and coordination via intrinsic rewards.

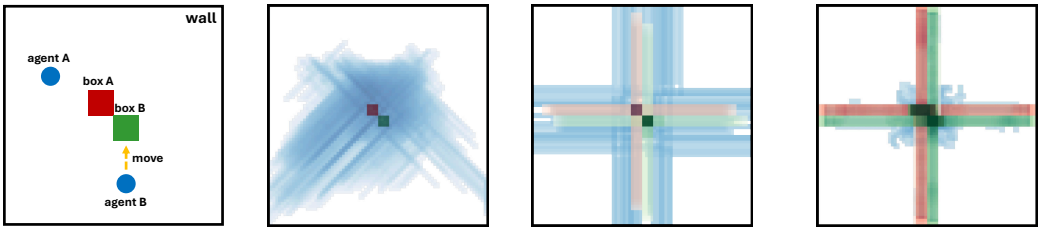
Eligibility Trace. Eligibility traces are used to implement TD(λ) online by propagating the current TD error to future timesteps for value updates [67]. At each timestep t, the trace $e_t(s)$ is updated as:

$$e_t(s) = \begin{cases} \gamma\lambda e_{t-1}(s) + 1, & \text{if } s = s_t, \\ \gamma\lambda e_{t-1}(s), & \text{otherwise}, \end{cases} \tag{2}$$

where λ is the decay factor. This mechanism accumulates eligibility for recently visited states and decays it over time, thereby concentrating value updates on recently visited states. In this work, we adapt this concept to sustain coordinated influence, encouraging multiple agents to persistently focus on under-explored state dimensions.

4 Methodology

4.1 The Need for Advanced Focusing Influence

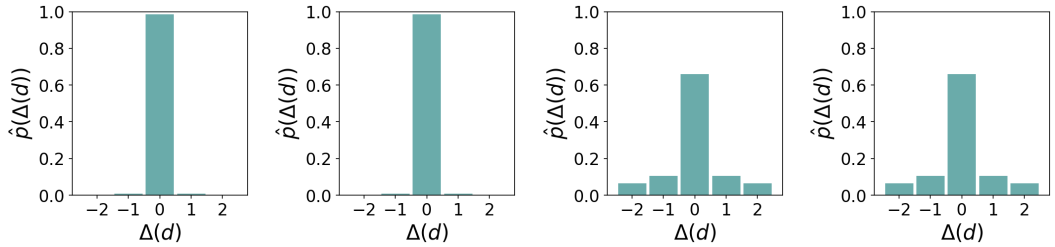


(a) Push-2-Box (b) Vanilla QMIX (c) QMIX with AFI (d) QMIX with AFI and SFI (proposed FIM)

Figure 1: Comparative results in the Push-2-Box environment: (a) shows an enlarged view of the environment, and (b–d) show average visitation counts of two agents (blue) and two boxes (Box A: red, Box B: green) over 3M timesteps across 100 seeds. Darker areas indicate more frequent visits.

In cooperative MARL, agents must often discover coordinated behaviors through exploration, particularly in tasks where progress depends on cooperative joint actions [10–12]. While each agent individually possesses the capacity to influence the environment, sparse rewards provide insufficient learning signal to coordinate their influences. As a result, agents act in an uncoordinated manner with scattered, non-cooperative influence, failing to converge on shared targets. To illustrate, we consider the Push-2-Box environment [11] in Fig. 1(a), which involves two agents and two boxes. A single agent moves a box by one cell, while simultaneous pushing moves it by two cells; agents must coordinate to reach a wall within the episode time limit. Positive rewards are provided only when a box reaches the wall, and thus success requires persistent, synchronized pushing of the same box. Under such reward sparsity, CTDE methods often fail to discover the need for coordinated pushing during training, resulting in limited variation in the box position and task failure. Fig. 1(b) illustrates this failure mode for QMIX, where scattered exploration and weak coordination prevent success.

These observations motivate the need to (i) focus multi-agent influence on shared targets, and (ii) prioritize under-explored state dimensions under the current behavior policy. To this end, we propose the **Focusing Influence Mechanism (FIM)**, which promotes cooperation through two components: *agent focusing influence (AFI)* and *state focusing influence (SFI)*. AFI encourages agents to align their influence on the same targets, for example by focusing on the same box, by accumulating collective



(a) Box A (QMIX) (b) Box B (QMIX) (c) Box A (with AFI) (d) Box B (with AFI)

Figure 2: Empirical distribution $\hat{p}$ of temporal changes $\Delta(d)$ for box positions in Push-2-Box. (a–b): Without AFI. (c–d): With AFI.

influence via eligibility traces and amplifying intrinsic rewards for dimensions associated with consistent joint attention, thereby driving coordinated behavior. As shown in Fig. 1(c), incorporating AFI into QMIX increases collective influence on shared targets; however, since AFI operates across all state dimensions, this influence can be spread too broadly, leaving insufficient effort on the dimensions that truly require coordination, such as box positions. SFI addresses this by weighting state dimensions using an entropy-based criterion, assigning higher priority to under-explored ones. Fig. 1(d) shows that combining AFI and SFI further concentrates joint influence on box positions, enabling successful task completion. Several prior methods also incorporate state influence or exploration objectives [14, 17, 30]. In Section 5, we compare these methods on Push-2-Box and show that, despite the simplicity of the task, they fail to solve it reliably under sparse rewards, whereas FIM succeeds by providing the coordination and persistence required in this setting.

4.2 Agent Focusing Influence

To enable coordinated behavior, agents must not only act but also exert *aligned influence* on shared state dimensions. We therefore quantify how much a joint action contributes to changes in each state dimension beyond what would arise without coordination. To this end, we adopt a counterfactual formulation [38] to measure the collective influence on dimension d at timestep t:

$$\mathrm{Inf}_t^d(s_t, a_t) = \sum_{i=0}^{n-1} \left\{ \left| \hat{s}_{t+1}^d(s_t, a_t) - s_t^d \right| - \mathbb{E}_{a_t^i \sim \rho} \left[\left| \hat{s}_{t+1}^d(s_t, a_t^i, a_t^{-i}) - s_t^d \right| \right] \right\}, \tag{3}$$

where $\hat{s}(\cdot)$ approximates the environment dynamics, and ρ denotes a random policy that removes the coordinated contribution of agent i. This formulation isolates the additional state change attributable to coordinated joint actions, rather than independent or uncoordinated behavior.

While Inf_t^d captures instantaneous coordination, effective cooperation requires *sustained* and temporally consistent influence. To account for temporal accumulation, we introduce an eligibility trace:

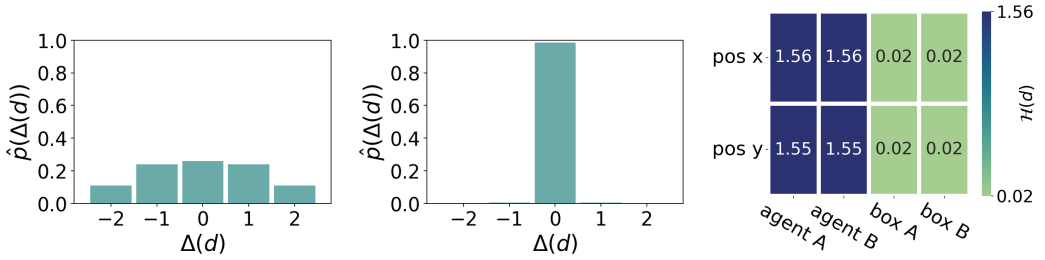
$$e_t^d = \gamma \cdot e_{t-1}^d + \mathrm{Inf}_t^d, \tag{4}$$

which aggregates past influence signals and reflects how consistently a dimension has been jointly affected over time. Using this trace, we define the **Agent Focusing Influence (AFI)** intrinsic reward:

$$r_t^{\mathrm{AFI}} = \sum_{d \in \mathcal{D}} \mathrm{Inf}_t^d \cdot \max(e_{t-1}^d, 1), \tag{5}$$

Here, $\mathcal{D} = \{0, \ldots, D-1\}$ indexes the state dimensions, and $\max(e_{t-1}^d, 1)$ keeps the trace multiplier at least one, while reinforcing dimensions that have received consistent collective attention. By weighting the current influence Inf_t^d with the accumulated trace e_{t-1}^d, AFI amplifies dimensions where coordinated influence is repeatedly exerted, encouraging agents to sustain coherent joint behavior rather than transient and unstructured interactions.

To illustrate the effect of AFI, Fig. 2 shows the empirical distribution of temporal changes in box positions in Push-2-Box, with and without AFI. Without AFI, panels (a) and (b) show that both boxes exhibit similarly low variation, reflecting the dispersed influence of vanilla QMIX.



(a) Agent positions (b) Box positions (c) Entropy $\mathcal{H}(d)$

Figure 3: (a–b) Empirical distribution $\hat{p}$ of temporal changes $\Delta(d)$ for agent and box positions. (c) Entropy $\mathcal{H}(d)$ for each state dimension in Push-2-Box.

With AFI, panels (c) and (d) show increased variation in both box positions, indicating stronger collective influence. However, since AFI aggregates influence across all state dimensions, this effect is distributed broadly, including agent positions that do not require coordination. As a result, insufficient influence is concentrated on box positions, limiting effective task completion. This limitation motivates a mechanism that selectively prioritizes which dimensions to focus on, which we address in Section 4.3.

4.3 State Focusing Influence and FIM Framework

To concentrate the coordinated influence induced by AFI on dimensions that matter for exploration, we introduce **State Focusing Influence (SFI)**, which identifies and prioritizes state dimensions that are under-explored under the current joint behavior policy $\boldsymbol{\beta}$. Our key observation is that under-explored dimensions exhibit low diversity in their induced transitions, which can be characterized by low conditional entropy $\mathbb{E}_{\boldsymbol{\beta}}\big[\mathcal{H}(s^d_{t+1} \mid s_t)\big]$. Intuitively, if a dimension rarely changes under the current policy, it receives little effective exploration, even over long horizons.

Proposition 4.1. *Let $\mathcal{E}_d := \sup_{t\geq 0} \mathcal{H}(s^d_{t+1} \mid s^d_t)$ be positive. To achieve a target exploration level H^d_{target} such that $\mathcal{H}(s^d_0, \ldots, s^d_T) \geq H^d_{\text{target}}$, the required horizon satisfies*

$$T \geq \frac{H^d_{\text{target}} - \mathcal{H}(s^d_0)}{\mathcal{E}_d}. \tag{6}$$

Proof. Proof of Proposition 4.1 is provided in Appendix A.

This result shows that when $\mathcal{H}(s^d_{t+1} \mid s^d_t)$ is small, each transition contributes little to exploration, requiring disproportionately long horizons. Therefore, such dimensions must be explicitly targeted to achieve sufficient coverage. However, directly comparing entropy across dimensions is inappropriate due to scale differences, since those with larger natural variation trivially yield higher entropy. To address this issue, we normalize each dimension by its expected change magnitude under $\boldsymbol{\beta}$:

$$\tilde{s}^d_t := \frac{s^d_t}{\mathbb{E}_{\boldsymbol{\beta}}[|s^d_{t+1} - s^d_t|] + \epsilon}, \tag{7}$$

which removes scale effects and enables fair comparison across dimensions, where ϵ is a small constant added for numerical stability.

We then approximate the conditional entropy using the marginal entropy of normalized changes:

$$\mathcal{H}(d) = \mathbb{E}_{\boldsymbol{\beta}}\big[-\log p\big(\Delta^d(s_t, s_{t+1})\big)\big], \tag{8}$$

where $\Delta^d := \tilde{s}^d_{t+1} - \tilde{s}^d_t$ and p is estimated empirically. This approximation captures how diverse the observed transitions are along each dimension under the current policy.

Based on this, SFI assigns higher importance to dimensions with lower entropy, i.e., those that are insufficiently explored. Integrating SFI with AFI yields the **Focusing Influence Mechanism (FIM)**:

$$r^{\text{FIM}}_t = \sum_{d\in\mathcal{D}} w_d \cdot \text{Inf}^d_t \cdot \max(e^d_{t-1}, 1), \tag{9}$$

where $w_d = \mathrm{Softmax}(-\mathcal{H}(d))$ emphasizes under-explored dimensions. To ensure stable adaptation to the evolving behavior policy, the weights w_d are updated using an exponential moving average:

$$w_d \leftarrow (1-\phi)\, w_d + \phi\, w_d^{\text{new}}, \tag{10}$$

where ϕ is the update rate. Crucially, FIM does not explicitly maximize entropy. Instead, it promotes exploration by allocating coordinated influence to under-explored dimensions, increasing their variability through joint behavior rather than stochasticity alone.

To visualize the effect of SFI, Fig. 3 shows the empirical distribution of temporal changes for agent and box positions, along with the corresponding entropy $\mathcal{H}(d)$ for each state dimension in Push-2-Box. Agent positions exhibit high entropy due to frequent changes, whereas box positions have low entropy because they require coordinated actions. Consequently, SFI assigns higher weight to box-position dimensions, directing AFI's influence toward targets that require joint effort.

Finally, the overall reward is defined as

$$r_t = r_{\text{ext},t} + \alpha \cdot r_t^{\text{FIM}},$$

where α is the intrinsic reward scale. We use QMIX [8] as the backbone, although FIM is agnostic to the underlying CTDE algorithm. The complete training procedure is summarized in Algorithm 1, with algorithmic details in Appendix B.1 and entropy-estimation settings in Appendix B.2.

5 Experiment

In this section, we evaluate the effectiveness of FIM. We begin with the Push-2-Box task introduced in Section 4.1, comparing various combinations of FIM components. We then extend the evaluation to more complex benchmarks, including the StarCraft Multi-Agent Challenge (SMAC) [68], SMACv2 [69], and Google Research Football (GRF) [70]. In all performance plots, the mean across 5 random seeds is shown as a solid line, and the standard deviation is represented by a shaded area.

5.1 Performance Comparison on Push-2-Box

We revisit the Push-2-Box task introduced in Fig. 1(a). In Push-2-Box, an external reward of $+100$ is given when either box reaches the wall, and -1 is applied if the task fails. The task is solved when agents push a box to the wall within the limited episode length, and detailed settings are provided in Appendix C. For comparison, Fig. 4 reports success rates for several MARL baselines: **LAIES** [17], which encourages influence over heuristic external state features; **CDS** [14], which promotes trajectory diversity; **FoX** [30], which leverages formation-aware exploration; and **vanilla QMIX**, trained using only extrinsic rewards. We also include component-wise variants: **QMIX+AFI**, which applies r_t^{AFI} with uniform weights across all state dimensions, and **QMIX+SFI**, which applies $\sum_{d\in\mathcal{D}} w_d \cdot \mathrm{Inf}_t^d$ without eligibility traces.

We observe that, despite the simplicity of the environment, only FIM solves the task. All other MARL baselines, including vanilla QMIX and exploration-oriented methods, fail on Push-2-Box. Notably, LAIES also fails even though it targets the same state dimensions in Push-2-Box, suggesting that identifying the correct dimensions is insufficient without AFI. Among our ablations, QMIX+AFI alone fails because concentrated influence without targeted dimension weighting is spread across all state dimensions, leaving insufficient joint effort on box positions to achieve the task, as shown in Fig. 1(c). QMIX+SFI increases interaction with hard-to-change components that require joint effort, but still fails due to inconsistent focus on a single target. Together, these results show that both SFI and AFI are necessary, and that their combination enables effective influence focusing.

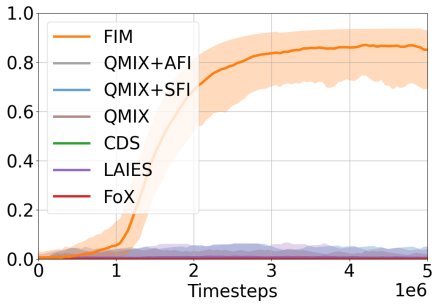


Figure 4: Performance comparison on the Push-2-Box environment.

5.2 Performance Comparison on Diverse Multi-Agent Environments

Next, we evaluate our method on three standard MARL benchmarks: SMAC, SMACv2, and GRF. SMAC is a multi-agent combat environment built on StarCraft II, where agents must coordinate to

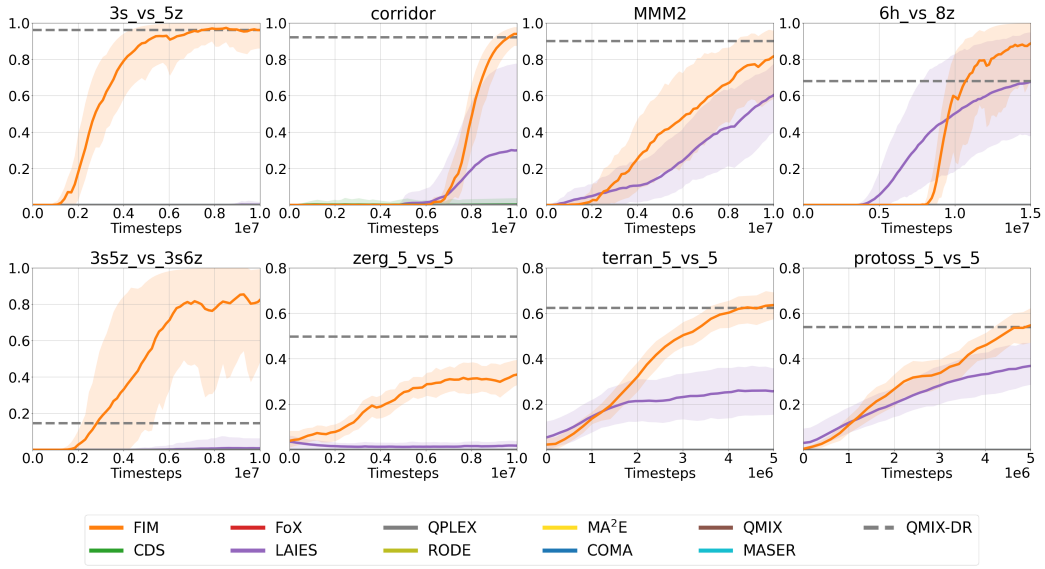


Figure 5: Performance comparison on SMAC and SMACv2 environments

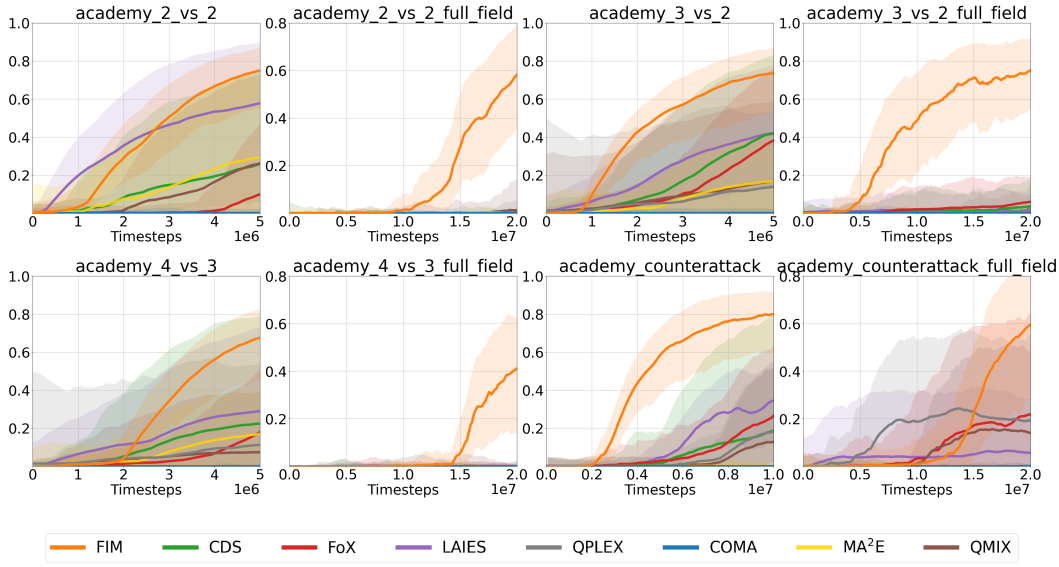


Figure 6: Performance comparison on GRF environments

defeat enemy units. We use a truly sparse reward setting in which agents receive +1 for a win, 0 for a draw, and -1 for a loss. Evaluation is conducted on five challenging scenarios: one hard map (`3s_vs_5z`) and four super-hard maps (`corridor`, `MMM2`, `6h_vs_8z`, `3s5z_vs_3s6z`), where `s`, `z`, and `h` refer to stalker, zealot, and hydralisk units, respectively. We further evaluate on SMACv2, which introduces randomized initial configurations and diverse unit types, using three scenarios (`zerg_5_vs_5`, `terran_5_vs_5`, `protoss_5_vs_5`) under the same sparse reward setting. GRF is a multi-agent soccer environment where teams compete to score goals under sparse rewards: +100 for a win and -1 for a loss. We evaluate on eight scenarios, including four half-field settings (`academy_2_vs_2`, `academy_3_vs_2`, `academy_4_vs_3`, `academy_counterattack`) and their corresponding full-field versions, which are more challenging due to the increased field size. Further environment details and visualizations are provided in Appendix C.

For SMAC and SMACv2, in addition to **Vanilla QMIX**, **LAIES**, **CDS**, and **FoX**, we include baselines compared with FIM variants: **COMA** [38], which uses counterfactual baselines for credit assignment; **MASER** [32], which identifies subgoals based on Q-values; **RODE** [71], which assigns roles to agents; **MA^{2}E** [72], which applies masking to reconstruct global information; and **QPLEX** [9],

which adopts a dueling architecture. For GRF, we compare against Vanilla QMIX, LAIES, CDS, FoX, COMA, MA^{2}E, and QPLEX, omitting methods without publicly available GRF results. We also include QMIX-DR for SMAC, trained with dense rewards, as a dense-reward reference rather than a strict upper bound. Its damage-based reward can bias learning toward distributed attacks, whereas the original sparse reward allows FIM to discover the focused-fire strategy that is particularly effective in `6h_vs_8z`. All baselines are evaluated using author-provided implementations, while our method uses the best hyperparameters identified via ablations. Further details, including algorithm descriptions and hyperparameter configurations, are provided in Appendix D.1 and Appendix D.2, respectively. Here, SFI in FIM typically assigns higher weights to opponent state dimensions that are difficult to change, whereas LAIES heuristically targets all enemy-related dimensions, highlighting a clear difference from our approach.

Fig. 5 and Fig. 6 present success rate comparisons on the SMAC, SMACv2, and GRF benchmarks. Across these environments, FIM consistently achieves the highest success rates among all baselines. In SMAC and SMACv2, sparse rewards pose a significant challenge because agents must eliminate all enemies without intermediate feedback. While LAIES remains competitive in some scenarios, it struggles on complex maps such as `3s_vs_5z` and `corridor`, since it treats all enemies as external state dimensions without focusing influence. In contrast, FIM remains robust by weighting under-explored dimensions. In GRF, although some baselines perform well on simpler half-field tasks, they largely fail on full-field maps with rare scoring opportunities. By concentrating influence on hard-to-change elements, FIM maintains strong performance across all scenarios. Overall, these results show that FIM promotes effective cooperation, enabling agents to solve challenging tasks even under highly sparse rewards.

To evaluate FIM beyond structured vector states, we consider Overcooked [73], whose spatial binary tensor encodes agents and ingredients, allowing FIM to identify under-explored dimensions such as the soup location. We also evaluate continuous actions in the Multi-Agent Particle Environment (MPE) [74, 75] with a MAPPO backbone. Both evaluations are detailed in Appendix E.1.

5.3 In-depth Analysis and Ablation Studies

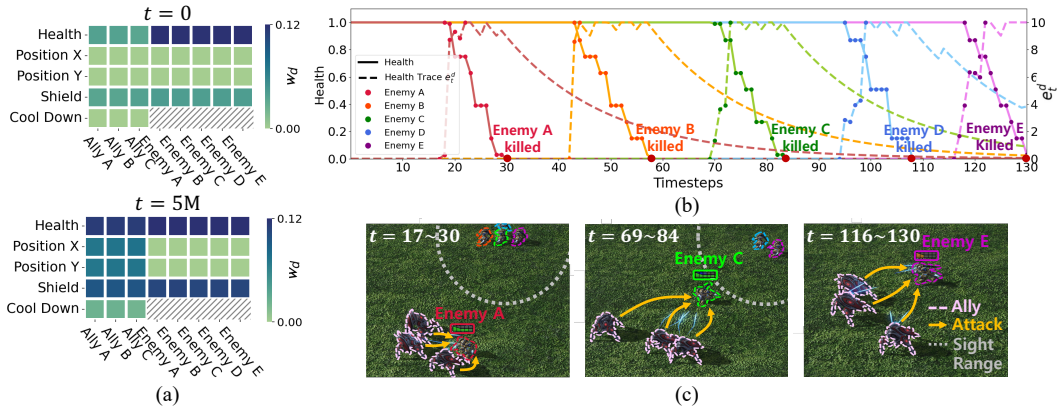
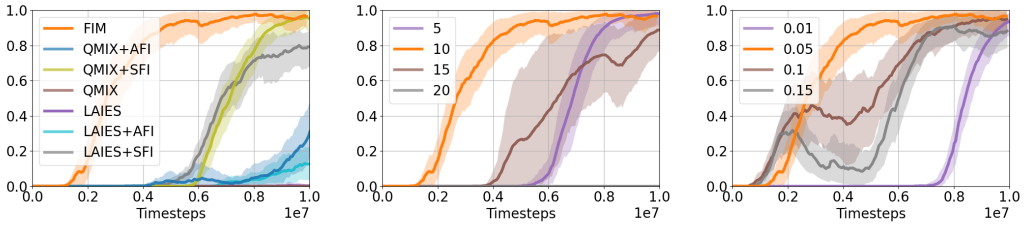


Figure 7: Behavior analysis in `3s_vs_5z`: (a) SFI weights w_d at 0 and 5M steps; larger weights prioritize low-entropy dimensions. (b) Changes in enemy health and its trace e_t^d. (c) Rendered frames for highlighting agents' coordination.

To better understand FIM's focusing mechanisms, we conduct detailed analyses and ablations in the environments where it shows the largest gains: SMAC `3s_vs_5z` and GRF `academy_3_vs_2_full_field`. In SMAC `3s_vs_5z`, SFI highlights enemy features such as health as under-explored state dimensions, as they remain relatively stable without coordination and thus become targets for joint influence. As shown in Appendix E.2, similar state dimensions exhibit low entropy in other SMAC scenarios. In GRF, the keeper's position is frequently up-weighted by SFI, since it is difficult for agents to manipulate. Overall, these results show that FIM focuses influence on meaningful state dimensions, improving performance by prioritizing under-explored features such as health in SMAC and the keeper's position in GRF.

Behavior Analysis: To further illustrate the effect of FIM, Fig. 7(a) reports the dimensional weights w_d across training iterations ($t = 0$ and 5M) in `3s_vs_5z`. Throughout training, the health of enemy units consistently receives high weights from SFI, since these dimensions change substantially only when agents coordinate their attacks to focus fire on the same target. Notably, as training progresses, ally health dimensions are also gradually up-weighted, since agents tend to avoid battle to preserve their own health. By assigning higher weight to ally health, SFI encourages agents to engage more actively with enemies. Fig. 7(b) shows how the corresponding eligibility traces evolve over the enemy-health dimensions, while Fig. 7(c) highlights key timesteps at which enemy units are eliminated within the trajectory. Agents trained with FIM learn to pull enemies into sight and focus fire sequentially: around $t \approx 20$, they concentrate on the first red enemy, increase its health trace, and eliminate it by $t \approx 30$; once removed, its influence drops to zero and attention shifts to the next enemy (e.g., orange at $t \approx 40$), repeating the same process. This strategy resembles human gameplay in StarCraft II. We provide a complementary analysis for GRF in Appendix F.1, showing that agents learn to disrupt the keeper's behavior, thereby increasing goal-scoring opportunities. Together, these results demonstrate that FIM promotes structured and effective cooperation even in sparse-reward environments.



(a) Component evaluation (b) Reward scale α (c) Update rate ϕ

Figure 8: Ablation studies on SMAC `3s_vs_5z`

Component Ablations: We evaluate the contribution of each component on `3s_vs_5z`. Fig. 8(a) compares **Vanilla QMIX**, **QMIX+AFI**, **QMIX+SFI**, **LAIES**, **LAIES+AFI**, which rescales LAIES's influence using per-dimension eligibility traces, **LAIES+SFI**, which applies w_d as a per-dimension weight on LAIES's influence signal, and **FIM**. The results show that Vanilla QMIX fails to solve the task, while SFI and AFI each provide consistent improvements regardless of the backbone: both QMIX+AFI and LAIES+AFI benefit from enforcing consistent focus on shared targets via eligibility traces, and both QMIX+SFI and LAIES+SFI benefit from concentrating influence on a small number of under-explored dimensions. Combining SFI and AFI yields faster convergence and higher final success rates, confirming that the two components play essential and synergistic roles in influence-based exploration. An ablation of the influence definition is provided in Appendix F.2.

Hyperparameter Analysis: Panels (b)-(c) of Fig. 8 examine the effects of the intrinsic reward scale α and dimension-weight update rate ϕ. For α, too-small values provide insufficient incentive for agents to concentrate their influence, while overly large values cause agents to over-prioritize intrinsic rewards over extrinsic feedback, leading to unstable training. In practice, α is environment-dependent, but can be effectively chosen by scaling it as large as possible without dominating the extrinsic reward signal. For ϕ, too-small values cause the dimension weights w_d to adapt too slowly to the evolving behavior policy, whereas too-large values result in frequent weight shifts that destabilize the focus target and prevent sustained coordinated influence. Although larger ϕ leads to faster adaptation, the weights eventually converge over time, suggesting that ϕ primarily controls the adaptation timescale rather than the final behavior. We use $\alpha = 10$ and $\phi = 0.05$ as default hyperparameters.

Additional analyses, including comparisons with alternative dimension-weighting strategies and ablations on more scenarios, are provided in Appendix F.2. The learned dynamics model is further analyzed in Appendix F.3, where we show that its evolving prediction error correlates with exploratory effectiveness and overall performance. Additionally, Appendix F.4 reports computational complexity, showing that FIM achieves superior performance with training time comparable to QMIX.

Table 1: Final win rates (mean ± std) on SMAC `3s5z_vs_3s6z`.

QMIX	QMIX + RND	QMIX + ICM	QMIX + Count	FIM (count)	FIM (frozen)	**FIM**
0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.04 ± 0.18	0.71 ± 0.27	$\mathbf{0.82 \pm 0.24}$

Table 2: Scalability on SMAC: final win rates and FIM training steps to convergence.

Scenario	Agents	LAIES	**FIM**	FIM steps to converge
`3s5z_vs_3s6z`	8	0.04 ± 0.05	$\mathbf{0.82 \pm 0.24}$	$\approx$ 7M
`3s6z_vs_3s7z`	9	0.00 ± 0.00	$\mathbf{0.80 \pm 0.25}$	$\approx$ 11M
`3s7z_vs_3s8z`	10	0.00 ± 0.00	$\mathbf{0.79 \pm 0.27}$	$\approx$ 25M

Entropy Criterion and Curiosity Controls. Low transition entropy can reflect strong control, but also a limited range of observed transitions; broader coverage can help escape local optima. We isolate focused influence from curiosity using FIM with a pre-trained, frozen dynamics model and QMIX + RND [76], ICM [77], or Count [78], with intrinsic rewards scaled to avoid dominating extrinsic feedback. FIM(count) replaces entropy-based weights with a count-based metric. On `3s5z_vs_3s6z`, updating the dynamics model improves on frozen dynamics, the curiosity baselines fail, and FIM(count) performs substantially worse (Table 1). These results support focused influence and entropy-based selection. Entropy-surrogate comparisons and estimation sensitivity are analyzed in Appendices F.5 and F.6.

Scalability. As teams grow, FIM must evaluate more agent-specific counterfactuals and prioritize influence over a larger state representation. We therefore test whether its focusing mechanisms remain effective as the coordination problem expands. Starting from the super-hard SMAC map `3s5z_vs_3s6z`, we increase the allied team from 8 to 10 agents and the opposing team from 9 to 11 units. Table 2 shows that FIM maintains final win rates of 0.82, 0.80, and 0.79, whereas convergence requires approximately 7M, 11M, and 25M steps. Thus, FIM preserves final performance in these larger scenarios, while the growing interaction and state spaces increase the training required.

6 Limitation and Future Work

Although FIM achieves strong performance, it still has some limitations. FIM incurs additional computational cost due to training the dynamics model, as analyzed in Appendix F.4. However, our analysis shows that this overhead is moderate relative to the performance gains, and it could be further reduced by using more lightweight model approximations. Second, although we validate FIM with spatial binary tensor states through Overcooked experiments, as shown in Appendix E.1, it is not yet specifically optimized for image-based environments. In such settings, learning to identify and focus on semantically meaningful regions or objects in the observation space could further improve performance. Deeper integration with image-based representations, for example through object-centric or detection-based mechanisms, remains an important direction for future work.

7 Conclusion

In this paper, we address the challenge of efficient cooperation in sparse-reward MARL by proposing FIM, a framework that guides agent influence toward under-explored state dimensions and sustains coordinated focus through eligibility traces. By integrating entropy-based dimension weighting with structured intrinsic rewards based on counterfactual reasoning, FIM enables agents to induce targeted and persistent state transitions. Empirical results across Push-2-Box, SMAC, SMACv2, and GRF demonstrate that FIM significantly improves learning efficiency and coordination, outperforming state-of-the-art baselines. These findings highlight the potential of influence-guided learning to enable robust multi-agent cooperation in sparsely rewarded environments.

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A Proof of Proposition 4.1

Proof. By the chain rule of entropy:

$$\mathcal{H}(s_0^d, s_1^d, \ldots, s_T^d) = \mathcal{H}(s_0^d) + \sum_{t=0}^{T-1} \mathcal{H}(s_{t+1}^d \mid s_0^d, \ldots, s_t^d) \tag{11}$$

$$\leq \mathcal{H}(s_0^d) + \sum_{t=0}^{T-1} \mathcal{H}(s_{t+1}^d \mid s_t^d) \tag{12}$$

$$\leq \mathcal{H}(s_0^d) + T \cdot \mathcal{E}_d, \tag{13}$$

where the first inequality follows from the fact that conditioning reduces entropy, and the second from the definition of $\mathcal{E}_d := \sup_{t \geq 0} \mathcal{H}(s_{t+1}^d \mid s_t^d)$. Therefore, to achieve $\mathcal{H}(s_0^d, s_1^d, \ldots, s_T^d) \geq H_{\text{target}}^d$, we need:

$$\mathcal{H}(s_0^d) + T \cdot \mathcal{E}_d \geq H_{\text{target}}^d, \tag{14}$$

which gives:

$$T \geq \frac{H_{\text{target}}^d - \mathcal{H}(s_0^d)}{\mathcal{E}_d}. \tag{15}$$

□

B Implementation Details

We first present the algorithmic details in Appendix B.1, followed by the entropy-estimation settings in Appendix B.2.

B.1 Algorithmic Details of FIM

Our implementation is available at `https://unistmlic.github.io/FIM/`.

Computing the influence Inf_t^d in Eq. (3) relies on predicting next-state outcomes. The dynamics model is initialized randomly and trained online together with the policy. FIM implements the transition model $\hat{s}$ as a three-layer MLP and minimizes the mean squared error:

$$\mathcal{L}_{\hat{s}} = \mathbb{E}_{s_t, \mathbf{a}_t, s_{t+1}} \left[\left\| \hat{s}(s_t, \mathbf{a}_t) - s_{t+1} \right\|^2 \right] \tag{16}$$

The FIM framework follows the CTDE paradigm, using QMIX to learn a joint action-value function. Each agent i has a Q-function $Q^i(\tau_t^i, a_t^i)$ based on its action-observation history τ_t^i and action a_t^i. These per-agent utilities are combined via a mixing network to produce a global joint Q-value, $Q_\theta^{\text{tot}}(s_t, \mathbf{a}_t)$, where θ denotes the parameters of the agent and mixing networks. To stabilize learning, FIM employs a target mixing network $Q_{\theta^-}^{\text{tot}}$, which is periodically updated by overwriting its parameters. The temporal difference (TD) loss is computed using a Bellman update:

$$\mathcal{L}_{\text{TD}}(\theta) = \mathbb{E}_{s, \mathbf{a}, r, s'} \left[\left(r_t + \gamma \max_{\mathbf{a}'} Q_{\theta^-}^{\text{tot}}(s_{t+1}, \mathbf{a}') - Q_\theta^{\text{tot}}(s_t, \mathbf{a}_t) \right)^2 \right] \tag{17}$$

The parameters θ are updated using the Adam optimizer, while the target network parameters θ^- are synchronized at fixed intervals. The complete training procedure of FIM is summarized in Algorithm 1.

Algorithm 1 FIM framework

Initialize: Q network, dynamics model $\hat{s}$
Collect trajectories under behavior policy
Approximate $\mathcal{H}(d)$ with the obtained trajectories based on Eq. (8)
Compute w_d for each $d \in \mathcal{D}$
for each training iteration **do**
  for each timestep t **do**
    Sample transition $(s_t, \mathbf{o}_t, \mathbf{a}_t, s_{t+1}, \mathbf{o}_{t+1})$ using $\boldsymbol{\pi}$,
      where $\mathbf{o}_t = (o_t^0, \cdots, o_t^{n-1})$
    for each $d \in \mathcal{D}$ **do**
      Compute influence Inf_t^d by Eq. (3)
      Update eligibility trace e_t^d by Eq. (4)
    end for
    Compute intrinsic reward r_t^{FIM} by Eq. (9)
  end for
  Update value function Q^{tot} and dynamics model $\hat{s}$
  if interval of 0.5M timesteps **then**
    Re-estimate $\mathcal{H}(d)$ under current behavior policy based on Eq. (8)
    Update w_d for each $d \in \mathcal{D}$ by Eq. (10)
  end if
end for

B.2 Empirical Estimation of $\mathcal{H}(d)$

To estimate $\mathcal{H}(d)$, we construct the empirical distribution $\hat{p}\left(\Delta^d(s_t, s_{t+1})\right)$ by counting occurrences discretized to two decimal places from 100K episodes collected under the current behavior policy. The entropy is then computed as:

$$\mathcal{H}(d) \approx \mathbb{E}_{\boldsymbol{\beta}}\left[-\log \hat{p}(\Delta^d(s_t, s_{t+1}))\right] \tag{18}$$

To ensure comparability across environments, $\mathcal{H}(d)$ values are min-max normalized to the range $[0, 1]$ within each environment.

C Environment Details

Push-2-Box: Push-2-Box is a cooperative multi-agent environment where two agents must jointly push one of two boxes toward a wall to obtain a reward, as depicted in Fig. 9. A box moves one cell if pushed by a single agent and two cells if pushed simultaneously. Thus, synchronized cooperation is essential for completing the task within the episode time limit. The environment terminates either when a box reaches the wall or when the episode length is exceeded. The **state space** consists of the (x, y) positions of agents and boxes, resulting in an 8-dimensional state vector. Each agent receives the full environment state as its observation. The **action space** is discrete, consisting of eight movement actions corresponding to up, down, left, right, top-right, right-down, down-left, and left-top directions. The **reward function** is described in Table 3.

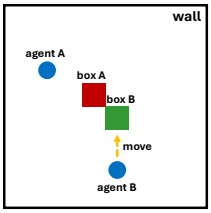


Figure 9: Push-2-Box

Table 3: Reward setting in Push-2-Box

Event	Reward
At least one box reaching the wall	+100 at episode end
No box reaching the wall	-1 at episode end

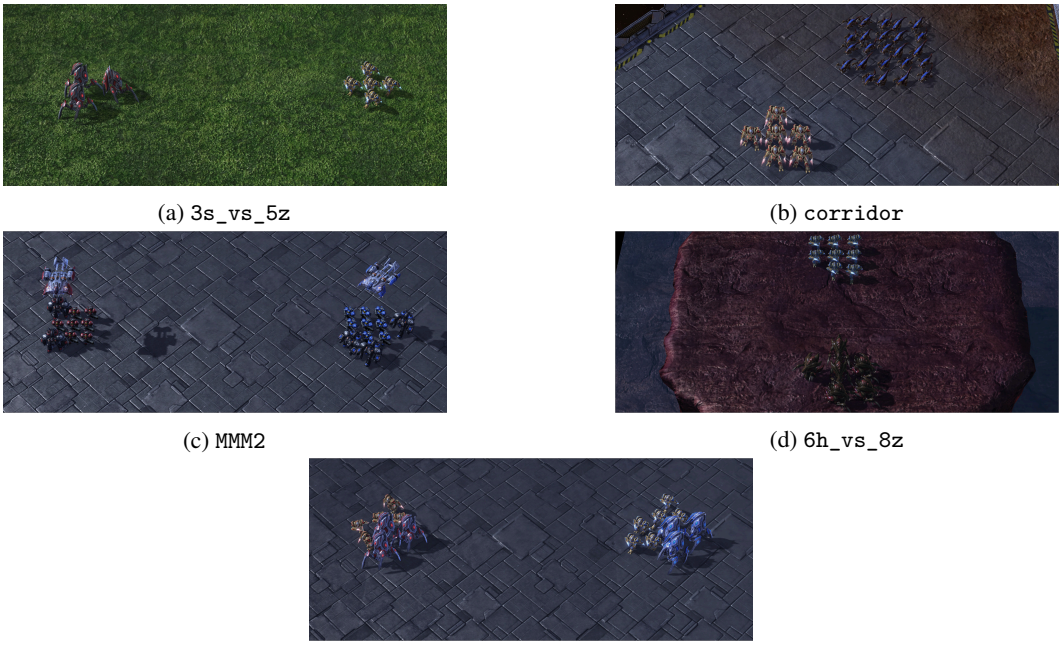
StarCraft Multi-Agent Challenge (SMAC & SMACv2): SMAC [68] is a benchmark for evaluating cooperative multi-agent reinforcement learning. Agents control individual StarCraft II units and coordinate to defeat enemy forces. Scenario visualizations are shown in Fig. 10, and unit compositions are summarized in Table 4. The **state space** contains global information about all units (e.g., positions, health, and shields), while the **observation space** provides each agent with local features of nearby units, including relative (x, y) positions and other attributes. The **action space** is discrete and consists of movement in four directions, attacking visible enemies, stopping, and a no-op action (only available to dead units). The **reward function** is summarized in Table 5, and our experiments focus on the sparse reward setting. We further evaluate on SMACv2 [69], an improved variant that introduces randomized unit compositions and initial positions, making coordination more challenging. The state, observation, and action spaces follow the same structure as SMAC, and we evaluate on three scenarios (`zerg_5_vs_5`, `terran_5_vs_5`, `protoss_5_vs_5`), with unit compositions provided in Table 4.

Table 4: SMAC and SMACv2 scenario configuration

Scenario	Ally	Enemy	State Dim	Obs Dim	Action Dim
`3s_vs_5z`	3 Stalkers	5 Zealots	68	48	11
`corridor`	6 Zealots	24 Zerglings	282	156	30
`MMM2`	1 Medivac, 2 Marauders, 7 Marines	1 Medivac, 3 Marauders, 8 Marines	322	176	18
`6h_vs_8z`	6 Hydralisks	8 Zealots	140	78	14
`3s5z_vs_3s6z`	3 Stalkers, 5 Zealots	3 Stalkers, 6 Zealots	230	136	15
`zerg_5_vs_5`	5 Zerg units	5 Zerg units	65	82	11
`terran_5_vs_5`	5 Terran units	5 Terran units	65	82	11
`protoss_5_vs_5`	5 Protoss units	5 Protoss units	75	92	11

Table 5: Sparse reward setting in SMAC and SMACv2

Event	Reward
Winning the battle	+1 at episode end
Losing the battle	-1 at episode end



(a) 3s_vs_5z (b) corridor

(c) MMM2 (d) 6h_vs_8z

(e) 3s5z_vs_3s6z

Figure 10: Visualization of SMAC scenarios.



(a) zerg_5_vs_5 (b) protoss_5_vs_5 (c) terran_5_vs_5

Figure 11: Visualization SMACv2 scenarios.

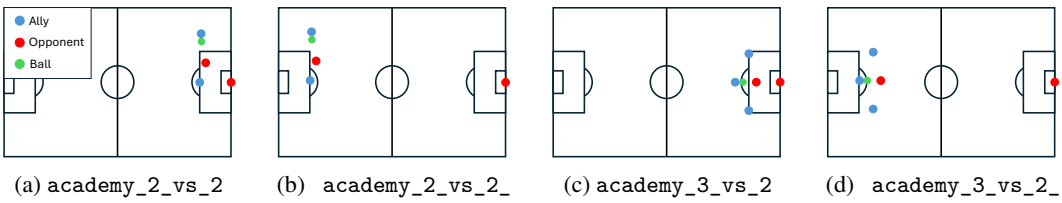
Google Research Football (GRF): GRF [70] provides a soccer simulation, in which agents must cooperate to score goals against an opponent team controlled by a scripted AI. We adopt a sparse reward setting to evaluate cooperative behavior under severely limited feedback. The **state space** consists of player positions and velocities, as well as ball position and velocity. Each **observation space** for an agent includes local information about the ego player, other players, and ball-related features, all expressed relative to the agent's current frame. The **action space** is discrete, covering movement in eight directions, sliding, passing, shooting, sprinting, and standing still. The **reward function** is described in Table 6. For brevity, several GRF scenarios are referred to using abbreviated names: `academy_3_vs_2` refers to `academy_3_vs_1_with_keeper`, `academy_2_vs_2` to `academy_run_pass_and_shoot_with_keeper`, `academy_counterattack` to `academy_counterattack_hard`, and `academy_4_vs_3` to `academy_4_vs_2_with_keeper` in the original GRF environment. As shown in Fig. 12, we design full-field variants of these scenarios by repositioning players to opposite half of the court. Table 7 provides an overview of the unit configurations.

Table 6: Reward setting in GRF

Event	Reward
Winning the match	+100 at episode end
Losing the match	-1 at episode end

Table 7: GRF scenario configuration

Scenario	Ally	Opponent	State Dim	Obs Dim	Action Dim
academy_2_vs_2	2 center back	1 goalkeeper, 1 center back	22	22	19
academy_2_vs_2 _full_field	2 center back	1 goalkeeper, 1 center back	22	22	19
academy_3_vs_2	3 central midfield	1 goalkeeper, 1 center back	26	26	19
academy_3_vs_2 _full_field	3 central midfield	1 goalkeeper, 1 center back	26	26	19
academy_4_vs_3	4 central midfield	1 goalkeeper, 2 center back	34	34	19
academy_4_vs_3 _full_field	4 central midfield	1 goalkeeper, 2 center back	34	34	19
academy _counterattack	1 central midfield, 1 left midfield, 1 right midfield, 1 central front	1 goalkeeper, 2 center back	34	34	19
academy _counterattack _full_field	1 central midfield, 1 left midfield, 1 right midfield, 1 central front	1 goalkeeper, 2 center back	34	34	19



(a) academy_2_vs_2 (b) academy_2_vs_2_full_field (c) academy_3_vs_2 (d) academy_3_vs_2_full_field

(e) academy_4_vs_3 (f) academy_4_vs_3_full_field (g) academy_counter attack (h) academy_counter attack_full_field

Figure 12: Visualization of initial agent positions in GRF scenarios.

D Experimental Details

FIM is implemented on top of the open-source framework from [79], which is also used to run QMIX [8] and QPLEX [9]. LAIES [17], RODE [71], MASER [32], CDS [14], and FoX [30] are evaluated using the original code and settings provided by their respective authors. Experiments are conducted on an NVIDIA RTX 3090 GPU with an Intel Xeon Gold 6348 CPU (Ubuntu 20.04). For the original benchmarks, training completes within two days for Push-2-Box and SMAC, while each GRF scenario requires less than two days to reach 5 million timesteps. We begin by describing the baseline algorithms in Appendix D.1 and outline the hyperparameter setup of FIM in Appendix D.2.

D.1 Detailed Description of Baseline Algorithms

- **QMIX** [8] factorizes the joint action-value into individual utilities combined by a monotonic mixing network. Code: `https://github.com/hijkzzz/pymarl2`
- **QPLEX** [9] extends QMIX with a duplex dueling architecture, decomposing joint value into individual value and advantage while enforcing the IGM principle. Code: `https://github.com/hijkzzz/pymarl2`
- **LAIES** [17] incentivizes agents to influence external task-relevant states via intrinsic rewards for both individual and joint impacts. Code: `https://github.com/liuboyin/LAIES`
- **RODE** [71] employs hierarchical role-based policies where agents periodically select roles to guide low-level actions, enabling scalable specialization. Code: `https://github.com/TonghanWang/RODE`
- **MASER** [32] explores by assigning subgoals from past trajectories, rewarding agents for revisiting informative states. Code: `https://github.com/Jiwonjeon9603/MASER`
- **CDS** [14] encourages policy diversity by maximizing mutual information between agent identity and trajectory. Code: `https://github.com/lich14/CDS`
- **FoX** [30] promotes exploration by maximizing entropy of agent formations and their mutual information with team structure. Code: `https://github.com/hyeon1996/FoX`
- **COMA** [38] is a multi-agent policy gradient method that assigns credit using a centralized critic with counterfactual baselines. Code: `https://github.com/oxwhirl/pymarl`
- **MA^{2}E** [72] addresses partial observability with a masked autoencoder to infer global information from individual observations. Code: `https://github.com/cheesebro329/MA2E`

D.2 Hyperparameter Setup of the Proposed FIM

Scenario-specific settings for the intrinsic reward weight α and update rate ϕ are provided in Table 9, while the softmax temperature in $\mathrm{Softmax}(-\mathcal{H}(d))$ is set to 0.1, and the discount factor γ is fixed at 0.99 across all scenarios. The default hyperparameter settings of FIM, which are generally shared across scenarios, are summarized in Table 8.

Table 8: Common hyperparameter settings of FIM

Hyperparameters	Value
Optimizer	Adam
Replay buffer size	5000
Target update interval	200
Mini-batch size	32
Mixing network dim	32
Discount factor γ	0.99
Learning rate	0.0005
Dynamics model $\hat{s}(\cdot)$ layers	3
Dynamics model $\hat{s}(\cdot)$ dim	128

Table 9: Scenario-specific hyperparameter setup of FIM

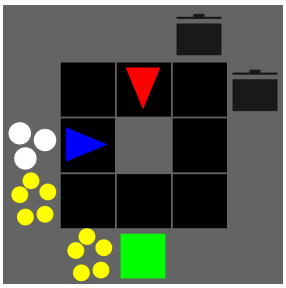
Scenario	α	ϕ
Push-2-Box	5	0.05
StarCraft Multi-agent Challenge (Sparse)		
3s_vs_5z	10	0.05
corridor	20	0.05
MMM2	10	0.05
6h_vs_8z	20	0.05
3s5z_vs_3s6z	10	0.05
StarCraft Multi-agent Challenge v2 (Sparse)		
protoss_5_vs_5	10	0.05
terran_5_vs_5	10	0.05
zerg_5_vs_5	10	0.05
Google Research Football (Sparse)		
academy_2_vs_2	10	0.05
academy_2_vs_2_full_field	10	0.05
academy_3_vs_2	10	0.05
academy_3_vs_2_full_field	10	0.05
academy_4_vs_3	10	0.05
academy_4_vs_3_full_field	10	0.05
academy_counterattack	10	0.05
academy_counterattack_full_field	10	0.05

E Additional Experiments

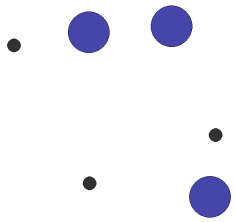
We evaluate alternative state representations and continuous actions in Appendix E.1, followed by SFI-weight visualizations in Appendix E.2.

E.1 Generalization Across State and Action Spaces

We test FIM beyond structured vector states and discrete-action value decomposition using spatial tensor states in Overcooked and continuous actions in MPE (Figure 13).



(a) Overcooked: `coord-ring`



(b) MPE: `simple_spread_v3`

Figure 13: Illustrative states of the additional environments. (a) Two agents prepare and deliver soup in a ring-shaped kitchen. (b) Agents (blue) cover landmarks (gray) while avoiding collisions.

Spatial tensor states in Overcooked. We evaluate Overcooked [73] `coord-ring` using the JaxMARL [80] implementation. Unlike structured representations that explicitly provide object coordinates, the state is a height $\times$ width $\times$ components tensor, with agents and ingredients encoded over a spatial grid. The soup location remains under-explored until preparation is complete. As training progresses, agents explore different positions with the soup and learn efficient goal-directed trajectories. With $\alpha = 1$ and $\phi = 0.05$, FIM improves IPPO's return (Table 10). The observations are spatial binary encodings, not raw images.

Continuous actions in MPE. We use the Multi-Agent Particle Environment (MPE) [74], specifically PettingZoo [75] `simple_spread_v3`, with continuous actions and a MAPPO [6] backbone. Agents coordinate to cover landmarks while avoiding collisions. Continuous actions preclude enumerating a finite action set for counterfactual evaluation. We therefore estimate the expectation in Eq. (3) by Monte Carlo sampling from the continuous policy: for each agent, we sample alternative actions while holding the other agents' actions fixed, and average the magnitudes of the predicted state changes. This preserves the influence-based reward and its SFI/AFI focusing mechanisms while changing how the counterfactual expectation is computed. After 1M training timesteps, FIM improves MAPPO's return (Table 10), supporting applicability to this continuous-action task. Reward scales differ across environments; comparisons are made within each setting.

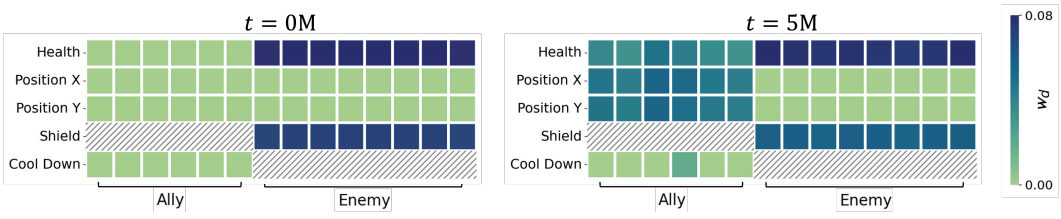
Table 10: Returns on Overcooked and MPE, comparing each backbone with and without FIM.

Method	Return
Overcooked — `coord-ring`	
IPPO	195.7 ± 5.3
IPPO + FIM (ours)	$\mathbf{218.5 \pm 4.7}$
MPE — `simple_spread_v3`	
MAPPO	-80.7 ± 19.6
MAPPO + FIM (ours)	$\mathbf{-63.4 \pm 7.9}$

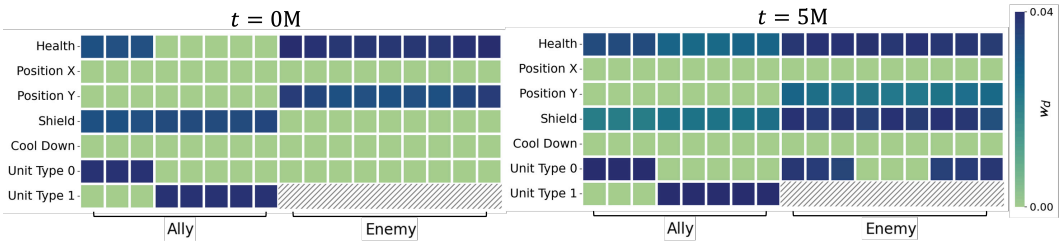
E.2 Visualization of SFI Weights w_d

Fig. 14 and Fig. 15 visualize w_d changes as training proceeds in SMAC and SMACv2, respectively. Across both benchmarks, SFI consistently assigns the highest weights to enemy health dimensions throughout training, reflecting their status as under-explored state dimensions that change substantially only when multiple agents coordinate their attacks on the same target. This makes enemy health a natural focal point for joint influence, as reducing it requires sustained and synchronized action from the team. Ally-specific features such as ally health also receive progressively elevated weights as training advances. This is a consequence of agents' tendency to avoid direct combat in order to preserve their own health, which causes ally health dimensions to remain relatively stable and thus under-explored. By up-weighting these dimensions, SFI counteracts this avoidance behavior and encourages agents to engage more actively in battle, ultimately promoting more effective coordination.

Fig. 16 visualizes w_d in GRF, where the opponent goalkeeper's position and direction dimensions are consistently assigned the highest weights throughout training. The goalkeeper's state exhibits persistently low entropy because it remains largely stationary under the joint behavior policy, shifting only in response to direct pressure from attacking agents. Consequently, SFI identifies these dimensions as primary targets, concentrating the team's cooperative effort on a state that is both stable and impactful. A detailed account of how this weight assignment translates into concrete agent behaviors is provided in Appendix F.1.

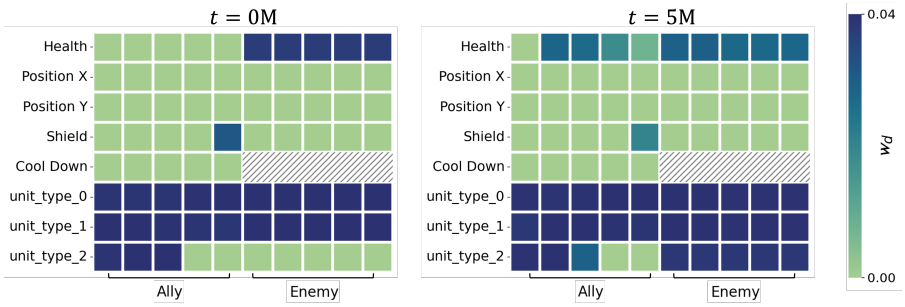


(a) `6h_vs_8z`

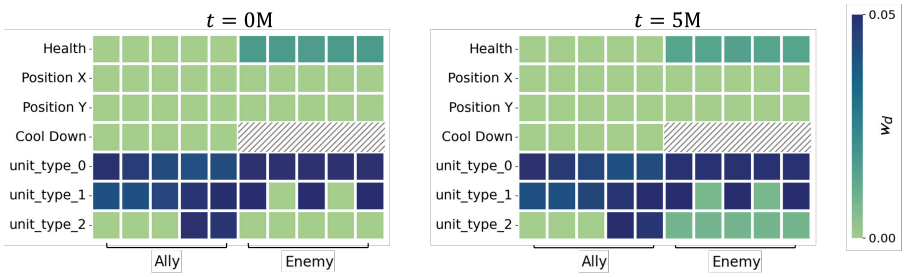


(b) `3s5z_vs_3s6z`

Figure 14: SMAC w_d visualization

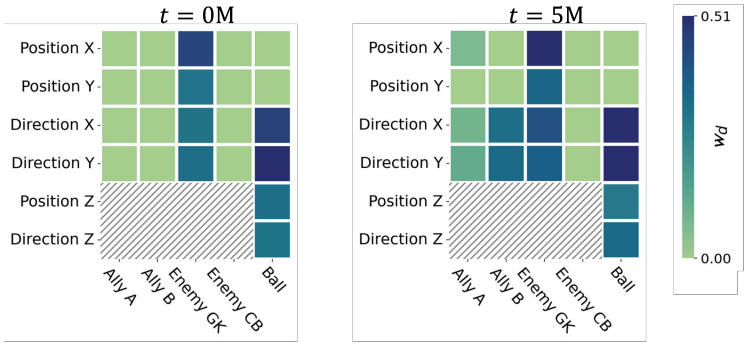


(a) `protoss_5_vs_5`

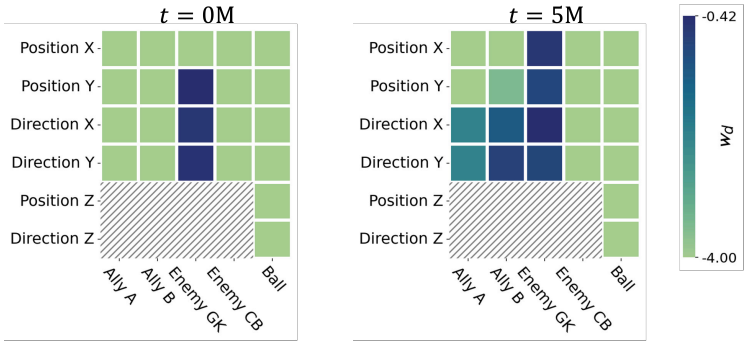


(b) `terran_5_vs_5`

Figure 15: SMACv2 w_d visualization



(a) `academy_2_vs_2`



(b) `academy_2_vs_2_full_field`

Figure 16: GRF w_d visualization

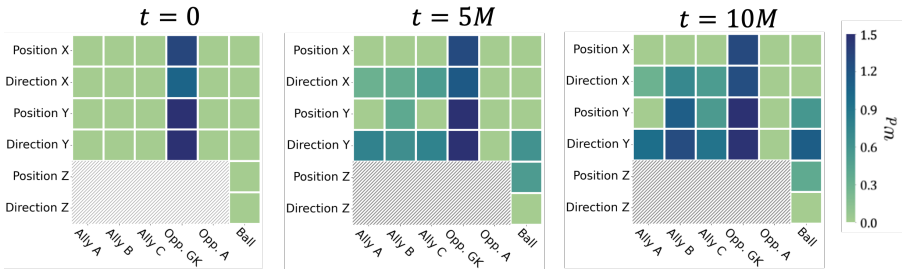
F Additional Analysis and Ablation Studies

We present GRF behavior analysis in Appendix F.1, extended ablations, including the influence-definition comparison, in Appendix F.2, dynamics-model analysis in Appendix F.3, computational costs in Appendix F.4, entropy-surrogate validation in Appendix F.5, and entropy-estimation sensitivity in Appendix F.6.

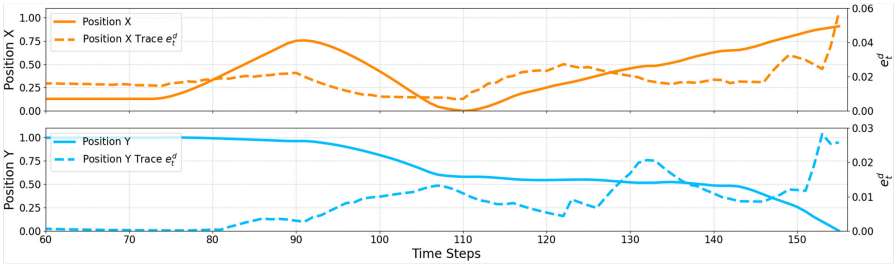
F.1 Strategy Analysis in GRF

In GRF, the dimension weights w_d consistently focus on the opponent goalkeeper's position and direction throughout training, as illustrated in Fig. 17(a). These dimensions exhibit low entropy throughout the training, since the goalkeeper typically remains stationary and only shifts position when a ball-carrying agent approaches the goalpost. This characteristic makes the goalkeeper's state both stable and strategically significant, as displacing it creates scoring opportunities and thus serves as a valuable proxy objective in sparse reward settings. Accordingly, FIM guides agents to influence the goalkeeper's position.

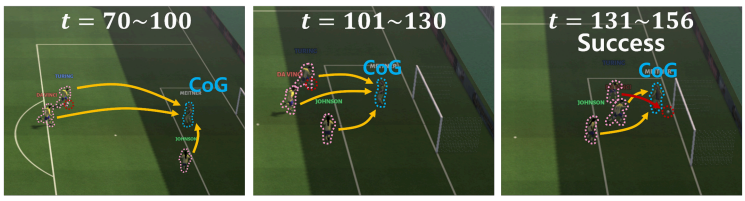
As shown in Fig. 17(b)-(c), this insight is reflected in the agent trajectory. Around $t \approx 70$ within the trajectory, agents begin to receive intrinsic rewards by subtly influencing the goalkeeper's position, even while positioned far from the goal area. By $t \approx 100$, the accumulated eligibility traces further incentivize agents to continue exerting influence over the goalkeeper, enabling a gradual progression toward the goal. Near $t \approx 130$, the goalkeeper briefly moves out of position, and the attacking agent capitalizes on this opportunity to score. Notably, FIM guides agents to approach the goal proactively and maintain persistent influence over the goalkeeper's positioning, which is an under-explored region under the joint behavior policy.



(a) Dimension weights w_d across iterations



(b) Changes in goalkeeper position and its trace e_t^d



(c) Rendered frames for highlighting agents' coordination

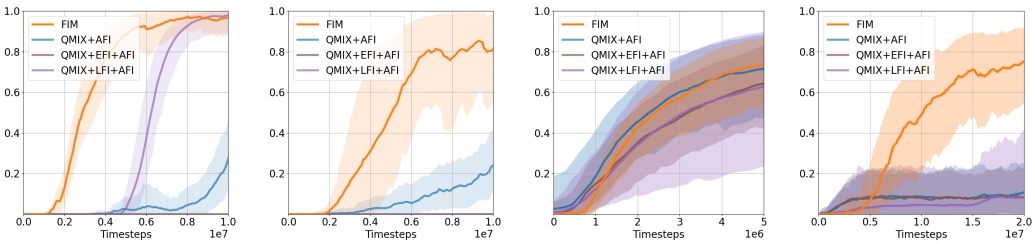
Figure 17: Analysis of FIM's strategy in GRF `academy_3_vs_2_full_field`

F.2 Extended Ablation Studies

To further evaluate the robustness of FIM, we conduct experiments on four challenging scenarios: SMAC `3s_vs_5z`, SMAC `3s5z_vs_3s6z`, GRF `academy_3_vs_2`, and GRF `academy_3_vs_2_full_field`, focusing on (i) alternatives to the SFI state focusing mechanism, (ii) the impact of each module in FIM through a component ablation study, (iii) the reward scaling factor α, and (iv) the update rate ϕ.

Alternatives to the SFI State Selection Mechanism: We investigate alternative strategies to the SFI state focusing mechanism for weighting state dimensions to influence: *uniform-weighting influence*, *external state focusing influence* (EFI), and *least-change state focusing influence* (LFI). In all cases, the intrinsic reward is computed as in FIM, with each variant differing only in weighting the state dimensions in $\mathcal{D}$. The uniform-weighting variant sets $w_d = 1$ for all $d \in \mathcal{D}$, which is equivalent to Eq. 5. EFI only weights external state features, following the approach of LAIES [17]: enemy health, shield, and positions in SMAC; and opponent and ball positions and directions in GRF. LFI assigns higher weights to state dimensions with smaller average temporal changes $|\tilde{s}^d_{t+1} - \tilde{s}^d_t|$ via softmax. In SMAC, this typically includes enemy health and ally positions, while in GRF, it often corresponds to ally direction features due to their relatively small-scale temporal changes.

As shown in Fig. 18, while some SFI variants show comparable performance in (a) and (c), the state dimensions weighted by SFI consistently lead to the highest overall performance. When variants include easily influenced features such as ally position, agents tend to exploit these trivial dimensions, leading to reward hacking and suboptimal behavior. These findings underscore the effectiveness of FIM's entropy-based weighting, which highlights stable and causally meaningful under-explored dimensions.



(a) `3s_vs_5z` (b) `3s5z_vs_3s6z` (c) `academy_3_vs_2` (d) `academy_3_vs_2_full_field`

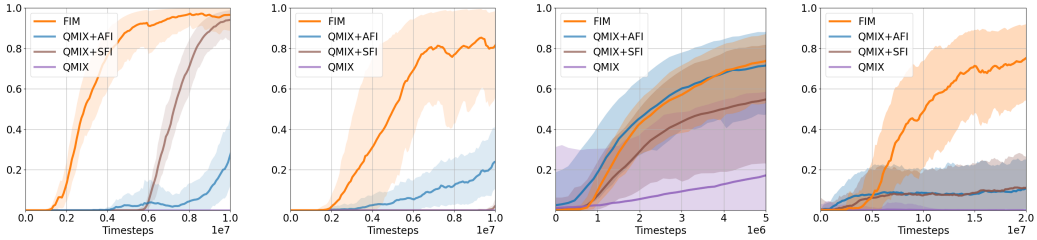
Figure 18: Alternatives to SFI

Effect of the Influence-Measure Definition. Complementing the state-selection alternatives above, we isolate the definition of influence by retaining FIM's SFI and AFI mechanisms and replacing only its influence measure with that of LAIES [17]. On SMACv2 `terran_5_vs_5`, this variant reaches 0.48 ± 0.12, compared with 0.59 ± 0.09 for FIM and 0.38 ± 0.14 for LAIES (Table 11). The comparison separates how influence is measured from how it is weighted and accumulated: the intermediate result supports a benefit from focusing even with the alternative measure, while FIM's state-change-magnitude definition provides a further gain in this experiment.

Table 11: Influence-definition ablation on SMACv2 `terran_5_vs_5`: final win rates.

Method	Win rate
LAIES	0.38 ± 0.14
FIM with the LAIES influence measure	0.48 ± 0.12
FIM (ours)	$\mathbf{0.59 \pm 0.09}$

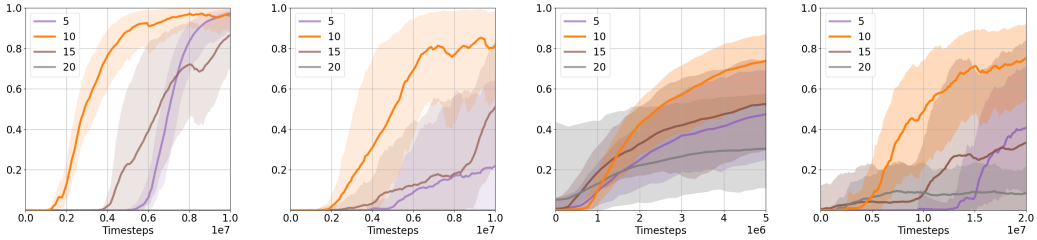
Component Evaluation: Fig. 19 compares four variants: vanilla QMIX, QMIX with state focusing influence (SFI), QMIX with agent focusing influence (AFI), and the full FIM framework that integrates both components. SFI improves performance by narrowing exploration to under-explored state dimensions, while AFI promotes deeper exploration by sustaining coordinated influence toward shared targets. Although each component contributes independently, only their combination yields the highest success rates.



(a) `3s_vs_5z` (b) `3s5z_vs_3s6z` (c) `academy_3_vs_2` (d) `academy_3_vs_2_full_field`

Figure 19: Component evaluation

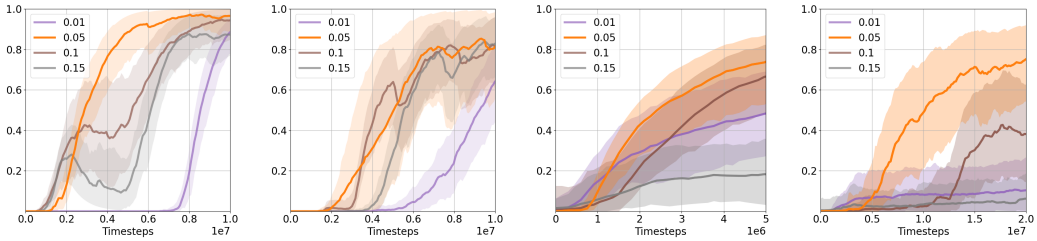
Reward scaling factor α: We examine how the reward scaling factor α affects performance by testing values in $\alpha \in \{5, 10, 15, 20\}$, as shown in Fig. 20. When α is too small, the intrinsic reward signal becomes negligible, preventing agents from effectively learning the influence strategy promoted by FIM. Conversely, setting α too large causes agents to over-prioritize intrinsic rewards, ignoring critical environmental feedback. Thus, we select $\alpha = 10$ as the default value that is well aligned with the extrinsic reward scale, guiding exploration without overwhelming the task objective.



(a) `3s_vs_5z` (b) `3s5z_vs_3s6z` (c) `academy_3_vs_2` (d) `academy_3_vs_2_full_field`

Figure 20: Reward scaling factor α

Update rate ϕ: We examine how the dimension-weight update rate ϕ affects performance by testing values in $\phi \in \{0.01, 0.05, 0.1, 0.15\}$, as shown in Fig. 21. When ϕ is too small, the dimension weights w_d adapt too slowly to the evolving behavior policy, failing to reflect which dimensions are currently under-explored and thus delaying effective coordination. Conversely, setting ϕ too large causes frequent weight shifts that destabilize the focus target, preventing agents from sustaining coordinated influence on any single dimension. Thus, we select $\phi = 0.05$ as the default, which strikes the best balance between stable and responsive adaptation to the current policy.



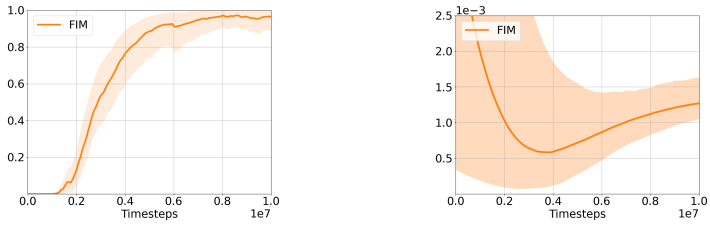
(a) `3s_vs_5z` (b) `3s5z_vs_3s6z` (c) `academy_3_vs_2` (d) `academy_3_vs_2_full_field`

Figure 21: Update rate ϕ

F.3 Analysis of the Learned Dynamics Model

Since the intrinsic reward in FIM is computed from the predictions of the learned dynamics model $\hat{s}$, its accuracy directly influences the reward signal. While a high mean-squared error (MSE) might seem detrimental, our results suggest that prediction inaccuracies can also serve a constructive role by implicitly encouraging exploration of regions with complex or less predictable dynamics. In this sense, model error may act as a form of curiosity, resonating with ideas from curiosity-driven exploration in model-based RL [77].

To examine this effect empirically, we analyzed the SMAC `3s_vs_5z` scenario. As shown in Fig. 22, the forward model's MSE initially decreased and then gradually increased during training, likely reflecting exposure to more diverse transitions. Notably, this trend coincided with a steady improvement in win rate, suggesting that moderate prediction error did not destabilize learning but rather correlated with productive exploration, ultimately supporting performance gains.



(a) FIM Performance (b) Mean squared error loss of $\hat{s}$

Figure 22: Comparison of FIM performance and mean squared error loss of $\hat{s}$ in `3s_vs_5z`

F.4 Comparison of Computational Complexity

FIM computes intrinsic rewards by estimating each agent's influence through counterfactual marginalization over its action set $\mathcal{A}$ for every dimension in $\mathcal{D}$. Storing all counterfactual state predictions requires $O(|\mathcal{N}| \cdot |\mathcal{A}| \cdot |\mathcal{D}|)$ memory per timestep. GPU batching reduces the wall-clock overhead of counterfactual evaluations, while the total arithmetic work still scales with the number of evaluations and state dimensions. FIM uses a lightweight three-layer multilayer perceptron (MLP) as the forward transition model and does not alter the main Q-network architecture, keeping computational overhead minimal. We compare FIM against QMIX with dense rewards (QMIX-DR). As shown in Table 12, FIM's average computation time per 1 million timesteps is comparable to QMIX-DR. In `3s5z_vs_3s6z`, FIM also requires fewer timesteps to reach a 60% success rate, demonstrating strong efficiency. These results show improved performance with modest wall-clock overhead in the evaluated scenarios.

Table 12: Computational cost: training time per 1M timesteps and steps to reach a 60% success rate.

Metric	QMIX-DR	**FIM (ours)**
SMAC — `3s_vs_5z`		
Time per 1M steps (min)	70.65	72.40
Steps to 60% success	4.21M	3.24M
SMAC — `3s5z_vs_3s6z`		
Time per 1M steps (min)	123.23	126.43
Steps to 60% success	13.05M	5.32M

F.5 Marginal and Conditional Entropy

SFI uses marginal transition entropy to rank state dimensions, whereas the theoretical motivation concerns entropy conditioned on the current state. We therefore compare $\mathcal{H}(\Delta^d)$ and $\mathcal{H}(\Delta^d \mid s_t)$, where $\Delta^d = \Delta^d(s_t, s_{t+1})$, on SMAC `3s5z_vs_3s6z` and GRF `academy_2_vs_2`. Conditioning can remove variation attributable to the current state, so these quantities need not be equal. We therefore examine whether the marginal surrogate preserves SFI's low-entropy ranking.

Table 13 shows numerically close estimates on both benchmarks. In SMAC, the dimension groups have the same ordering under both measures, with enemy health receiving the lowest entropy. In GRF, rankings are similar, and the low-entropy ball and goalkeeper features remain distinct from the more variable player features. These comparisons support the marginal surrogate in the evaluated settings; they do not establish equivalence for arbitrary state distributions.

Table 13: Marginal and conditional transition entropy for the listed feature groups. Shaded rows distinguish environments; GK denotes goalkeeper.

Feature group	Marginal $\mathcal{H}(\Delta^d)$	Conditional $\mathcal{H}(\Delta^d \mid s_t)$
SMAC — `3s5z_vs_3s6z`		
Enemy health	0.06	0.05
Enemy shield	0.61	0.57
Enemy position X	0.70	0.65
Enemy position Y	0.09	0.08
Ally health	0.40	0.37
Ally shield	0.12	0.10
Ally cooldown	0.88	0.83
Ally position X	0.74	0.69
Ally position Y	0.66	0.61
GRF — `academy_2_vs_2`		
Ally A position	0.73	0.72
Ally A direction	0.82	0.80
Ally B position	0.73	0.71
Ally B direction	0.78	0.73
Opponent GK position	0.27	0.21
Opponent GK direction	0.32	0.26
Opponent A position	0.74	0.70
Opponent A direction	0.96	0.94
Ball position	0.47	0.45
Ball direction	0.24	0.23

F.6 Sensitivity to Entropy Estimation

We examine whether SFI's preferred feature group depends strongly on the numerical choices used to estimate entropy. Starting from 2-decimal discretization, 100K episodes, and min–max normalization, we vary one factor at a time: discretization precision, the number of estimation episodes, or the normalization method. Discretization controls the resolution of empirical transition counts, the episode count controls the amount of estimation data, and normalization changes the scale on which transitions are compared.

Table 14 reports the top-weighted group at the start of training and after 5M timesteps on `3s5z_vs_3s6z`. Enemy health remains the highest-priority group for every tested setting at both checkpoints. This supports stable feature prioritization across these estimation choices. This comparison assesses feature-group selection at two checkpoints.

Table 14: Top-weighted state group under different entropy-estimation settings on SMAC `3s5z_vs_3s6z`. Only the named factor changes from the default.

Estimation setting	Top-weighted at 0M	Top-weighted at 5M
Default	Enemy health	Enemy health
1-decimal discretization	Enemy health	Enemy health
3-decimal discretization	Enemy health	Enemy health
50K estimation episodes	Enemy health	Enemy health
150K estimation episodes	Enemy health	Enemy health
No normalization	Enemy health	Enemy health
z-score normalization	Enemy health	Enemy health

G Broader Impact

This work advances cooperative multi-agent systems by introducing a framework that fosters coordinated behavior through influence-based intrinsic motivation. Enhanced cooperation among agents holds strong potential for positive societal impact in domains such as autonomous vehicle coordination, collaborative robotics, disaster response, and environmental monitoring. In these settings, the ability of agents to reason about and influence under-explored aspects collaboratively can lead to more robust, adaptive, and efficient team performance. As a foundational contribution, this research supports the development of AI systems that are better aligned with collective goals, promoting safer and more effective deployment in real-world multi-agent environments.

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